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Leprosy Diagnosis using Explainable Artificial Intelligence Techniques

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***Abstract*—Leprosy, also known as Hansen's Disease is a chronic curable infection that seldom causes skin lesion and nerve damage. The World Health Organization reported 127558 new leprosy cases detected globally in 2020. National Leprosy Eradication Program (NLEP) initiated by India, is the largest leprosy eradication program of the world. Despite this, 53.64% of leprosy cases (120,000 to 130,000 per year) are in India. Although this disease is curable at the later stages, an early diagnosis eradicates the complications incurred by nerve involvement. Existing research predicts leprosy on basis of electronic health records which require complex contact based techniques for procuration. Image based predictors lack credibility as it is hard to determine what features are being taken into consideration for prediction. This paper proposes an optimal AXI-CNN architecture for leprosy prediction called LeprosyNet. Deep Learning models are seldom black box in nature, in order to understand the area of interest in the images, explainable AI techniques like Activation Layer Visualization, Occlusion Sensitivity and Grad-CAM are used. The proposed model is compared with famous state-of-art architectures AlexNet and ResNet. Evaluation of the model has been undertaken using a ROC curve and confusion matrix. Accuracy obtained on LeprosyNet is 98%.**

Keywords—Explainable Artificial Intelligence, Deep Learning, Convolutional Neural Network, Gradient-weighted Class Activation Mapping (Grad-CAM), Occlusion Sensitivity Map.

I. INTRODUCTION

Leprosy stems from *Mycobacterium leprae*, an acid-fast, rod-shaped bacillus. This disease has a plethora of aftermaths like nerve injury, loss of sensation, other affected areas are the mucosa of the upper respiratory tract, and the eyes. Leprosy is curable and treatment in the early stages can prevent disability. According to a report by the World Health Organization, Hansen's disease is most prevalent in India, Brazil, Indonesia, Mozambique, Democratic Republic of the Congo.

Out of the above-mentioned countries, India accounts for the highest number of leprosy cases, closely followed by Brazil and Indonesia. The key to curing leprosy is early diagnosis. A delay in detection can have severe irreparable repercussions like crippling hands and feet, infertility, kidney failure, erectile dysfunction, paralysis, and blindness.

Despite the gravity of the issue, no prominent research has been done in the arena of computer vision based explainable lesion detection. The existing detection techniques require complex, expensive machinery that are contact based and invasive in nature. Deep Learning methods have paved their way in Disease detection due to high accuracy, easy deployment and their time efficient nature. Despite these advantages, unlike Machine Learning methods deep learning models do not work on set heuristics. Due to this, it becomes tedious to rely on their results in sensitive topics like disease prediction. In this paper, an optimal AXI tested CNN architecture called LeprosyNet has been proposed to make predictions, AXI techniques have been used to analyze important segments of the image on basis of which prediction is made. In order to test the robustness of the model in terms of credibility, transparency and interpretability, explainable AI techniques like visualization of activation layer, Occlusion Sensitivity and Grad-CAM have been used.

II. LITERATURE REVIEW

[1] clearly highlights that even though post-2005, notable changes were made in the National leprosy eradication program in India alongside global leprosy program, certain trends like detection of leprosy in children as well as disabilities caused due to delay in detection have not shown considerable regression. [5] Raises concern on the slow re-emergence of leprosy cases in India, whereas [6],[7] talk about permanent disabilities and sequelae, caused due to delay in diagnosis, thereby re-emphasizing the need for newer more efficient techniques of detection.

[11] & [12] Talk about the condition of leprosy in India, the government shed light on an important aftermath of NLEP, a program initiated to eliminate leprosy till 2005. Due to the lack of farsightedness of this program research regarding leprosy detection was put to a halt whereas new cases kept emerging and remained undetected. [13] States that a majority of cases originate from poverty stricken and impoverished regions of India, thus making affordable and readily accessible methods of detection an absolute necessity. [4] portrays the results of a demographic survey conducted on 187 patients between 1973 to 2018. A few noteworthy points of the study are, the median delay in diagnosis for patients is more than 3 years, and due to late detection patients developed significant comorbidities.

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In [14],[15] and [16] authors have shed light on the black-box nature of the predictions made by deep learning. Due to the lack of feature extraction steps in deep learning it is seldom hard to determine the underlying basis of prediction. In the arena of disease prediction, it is necessary to understand the working of the models before putting them on a pedestal because of high accuracy. In [17] and [18] authors have strived to combat COVID-19 by developing explainable detection models using LIME and SHAP techniques. The Authors in [20] have built a COVID detector using deep neural networks and explained its results using occlusion maps. They have further claimed that occlusion maps are simpler to implement, interpret and are more intuitive in nature when compared to Grad-CAM. In [19] explainable computer vision models have been built and compared to study Histopathology. It is evident from the results of the literature survey that the topic of Explainable AI is relatively new and unexplored. Even though several researchers have toiled to create ethical predictors for mainstream issues like COVID detection, diagnosis in persistent age-old problems like leprosy is still uncredible. Thus, paving the way for more reliable methods in this sphere. Existing methods of detection based on AI techniques either use features that require contact based and complex methods for procuration or use image based detectors which have less credibility. As an extension of literature survey the features, merits and demerits of existing techniques are discussed in section nine.

III. DATASET

The image dataset has been collected by churning out relevant images from the dermnetnz dataset from Kaggle as well as manual cleaning of results obtained from web scraping. A skewed data set has the potential of providing high accuracy yet its predictions are seldom unreliable. Hence equal number of positive and negative examples are a part of the data set. Figure 1 shows some sample images from the dataset, Top layer i.e (a), (b) and (c) consists of examples of leprosy and the bottom layer i.e (d), (e) and (f) consists of non-leprosy images.



(a) Leprosy (b) Leprosy (c) Leprosy



(d) Non Leprosy (e) Non leprosy (f) Non Leprosy

FIGURE 1.Top layer represents positive training examples and bottom layer consists of negative training examples.

IV. PROPOSED METHODOLOGY

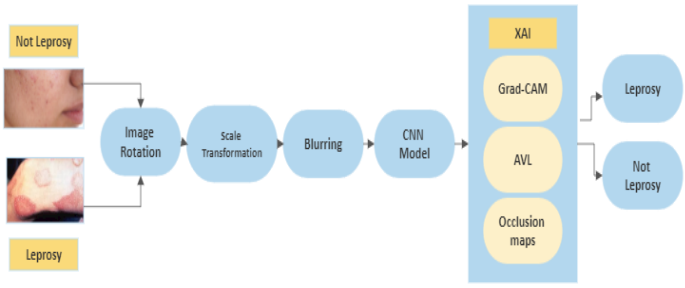


FIGURE 2. Architecture Flow

Fig.2 Depicts the proposed methodology of the paper. The input for the model is an image-based dataset consisting of two categories, leprosy and non-leprosy respectively. In the first phase of development image preprocessing techniques like rotation, scale transformations, and blurring have been undertaken on both testing as well as training subdivisions. After preprocessing, three CNN designs have been taken into consideration. Two are existing state-of-art architectures AlexNet and ResNet whereas the third is curated from scratch. To ensure the structural and prediction integrity of the most optimal CNN design three XAI techniques i.e., AVL, Grad-CAM and Occlusion Sensitivity Maps have been used. The final results are in the form of binary classification with the two categories being “Leprosy” and “Not Leprosy” respectively.

V. IMAGE PREPROCESSING

Some rudimentary geometric transformations in image preprocessing are undertaken in the initial stage of development for the sake of data augmentation as well as to make the detector robust in terms of change in scale, orientations, and camera properties. The standard image size before preprocessing was 1024 x 768 pixels. The size of the images only altered in scale transformation. The scaling has been done by a factor of 0.1 on both the x and y axis. Scale transformation has been achieved by using the built in resize function of OpenCV. The resultant image is thus downscaled and from Fig.3.a it is transformed to 3.c. Down scaled images not only contribute to data augmentation they are also easier to train due to low resolution. An angle between 0 to 90 degree is selected at random for each individual image in the dataset and rotation is done in accordance to that. The original image 3.a, after rotation changes to 3.b. Similarly, 3.a after blurring got converted to 3.d.

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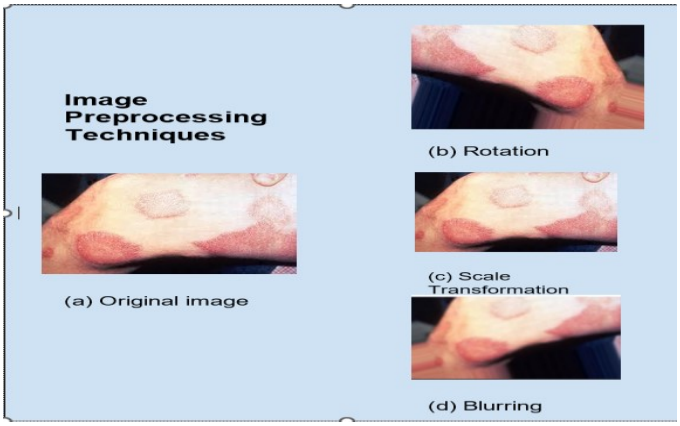


FIGURE 3. Dataset after image preprocessing

VI. LEPROSY NET ARCHITECTURE

One of the most eminent supervised learning algorithms in computer vision is Convolutional Neural Network. CNNs can reduce the high dimensionality of images without losing any information. Unlike the Multi-Layer Perceptron of Artificial Neural Networks where weights associated with every neuron are updated in an iteration of backpropagation, in CNN backpropagation only tends to update the weights of filters which then convolve on the entire image. This makes them computationally less expensive and the apt choice for image classification problems. Fig. 4 shows the architecture of the CNN used. The proposed model has a linear architecture consisting of four convolutional layers. The output of the second and fourth layer is max-pooled over a window of 3x3 to reduce the dimensionality of the inputs for the next convolutional layer. Batch Normalization is used between all layers to speed up training and use higher learning rates, making learning easier. Keras "padding= same" parameter is used in model which keeps the dimension of image the same as original after every convolution. ReLu has been used as an activation function in proposed architecture as it is significantly less prone to vanishing gradient problem in comparison to its other counterparts like sigmoid [9]. Due to its ability to fuse benefits of stochastic gradient descent as well as RMSProp, Adam has been chosen as the optimizer [21]. The performance of this architecture is compared with the existing state-of-the-art architectures like AlexNet and ResNet by extracting features from their fully connected layers and changing the classification layer in accordance with the problem.

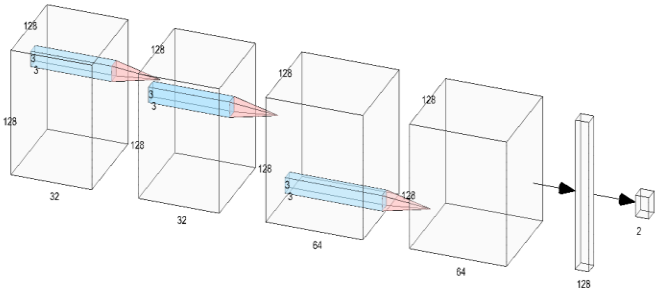


FIGURE 4. Architecture of the proposed CNN model (LeprosyNet)

VII. XAI (EXPLAINABLE ARTIFICIAL INTELLIGENCE) TECHNIQUES

(a) Activation Layer Visualization

This is one of the most rudimentary XAI techniques used in computer vision. The printing of all convolutional layers of a network can give us an insight on extracted features. The initial layers of any network are generic in nature and are preliminary used for edge detection. The job of the higher layers is to combine and build on the results of its predecessors. Fig 5. (a) Is the representation of the layers of LeprosyNet. The initial to final layer of the network is printed from top to bottom. Thus, the top layer is a generic template whereas in the bottom layer the template generated is specific and can be used to localize and identify lesion affected areas. Due to complexity of architecture of AlexNet and ResNet there is no clear demarcation between any two layers. All layers in Fig 5. (b) and (c) seem identical.

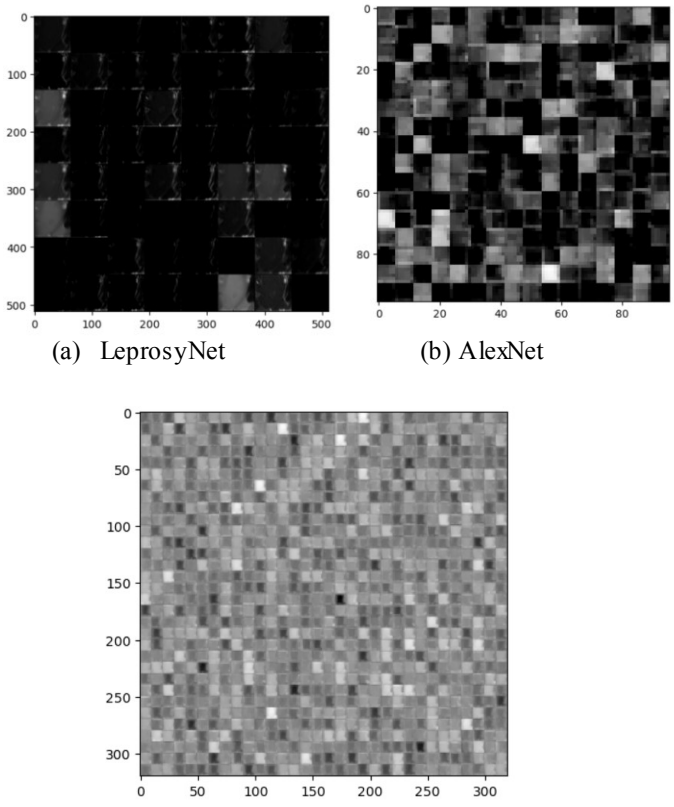


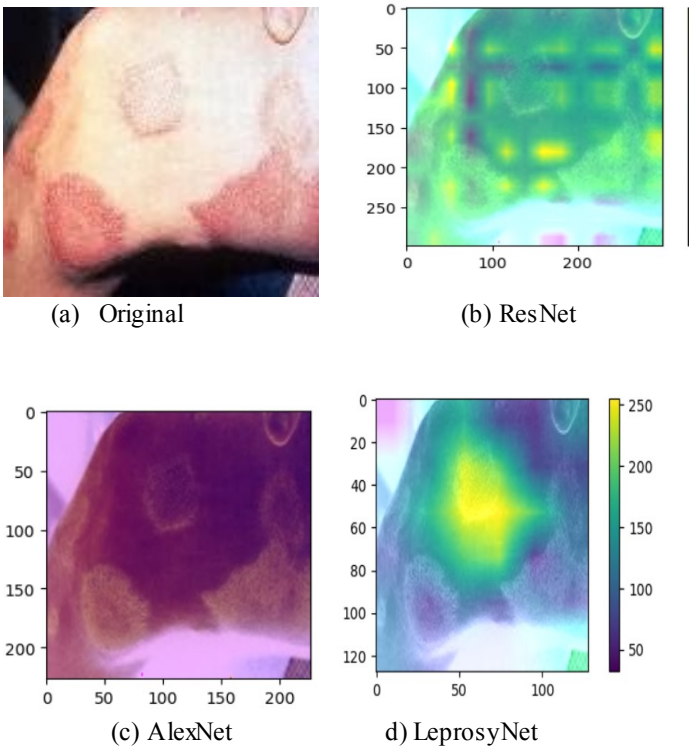
FIGURE 5. Visualization of Convolutional layers of proposed models

(b) Occlusion Sensitivity Map

In this technique a sensitivity map can be generated by placing a black box on different parts of the image and then analyzing how the occlusion affects the final probability at the time of classification. This can help us identify which areas of the image are more significant than the others for the neural network. Fig 6. Shows the results of Occlusion sensitivity maps obtained by applying the above-mentioned technique on three distinct models. Fig 6. (a) is the original image. One can easily analyze by the maps in image (b) and (c) that the models i.e.

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AlexNet and ResNet weren't successful in identifying the lesion afflicted areas. In (d) the LeprosyNet model has successfully identified the lesion affected area and there is a well-defined periphery between the redundant and affected spheres of the image.



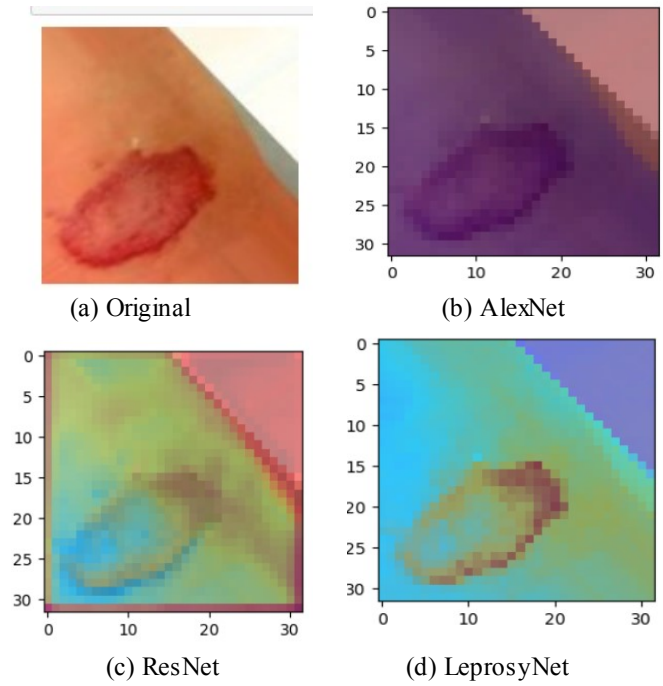
(a) Original (b) ResNet

(c) AlexNet d) LeprosyNet

FIGURE 6. Occlusion Sensitivity Maps.

(c) Grad-CAM

Gradient-weighted class activation mapping (Grad-CAM) is another method that is used to highlight the parts in an image that are most crucial for final classification. The technique takes the summation of gradients calculated between every pixel in an image and the feature map obtained by the last activation layer of the network. Fig. 7 shows the results obtained by using this technique on the proposed networks. Although Grad-CAM has also successfully highlighted the lesion affected areas, the results obtained are less intuitive than Occlusion maps. Grad-CAM are faster than Occlusion maps in terms of computation yet the spatial resolution of results is lower and some finer details remain unhighlighted. Fig. 7. (a) shows the original image, (b) is the Grad map of AlexNet and (c) ResNet. In both (b) and (c) the lesion affected area isn't highlighted whereas in Grad-Map of LeprosyNet (d), the corner pixels in the right side of the affected area are significantly darker than the rest of the image.



(a) Original (b) AlexNet

(c) ResNet (d) LeprosyNet

FIGURE 7. Grad-CAM map

VIII. Result and Discussion

The performance matrix used for evaluation of the proposed model is confusion matrix and ROC curve respectively. The dataset has been split into ratios of 80:20, 20% dataset is used for testing. The images in the test dataset are selected at random. Table 1 shows the results obtained on LeprosyNet, AlexNet and ResNet. The count of true positives, false negatives, false positives and true negatives has been obtained from the confusion matrix. Using the above data Precision (1), Recall (2), F1 (3) and Accuracy (4) have been calculated. The average accuracy of LeprosyNet is 0.98 which is better than the average accuracy of 0.81 obtained on AlexNet and 0.80 of ResNet. Fig. 8 shows the confusion matrix of LeprosyNet, Fig. 9 shows the confusion matrix of ResNet and Fig.10 AlexNet. ROC curves can be generated by plotting the true positive rate against the false positive rate. The ratio of correctly predicted positives out of all positive examples is known as the true positive rate (TPR), while false positive rate (FPR) is the number of incorrectly predicted examples out of all negative examples. This technique is used to gauge performance of binary classifiers. The threshold of the last sigmoid layer is changed to a value between 0-1 in every iteration and corresponding TPR and FPR are noted to generate the graph. Fig. 11 shows the ROC (receiver operating characteristics curve) for the three mentioned models, the AUC (area under curve) is highest for LeprosyNet which implies the quality of predictions made by the model are better than its counterparts. Architectures like AlexNet and ResNet are sophisticated architectures which can excel in generalization problems with plenty of categories. Linear and simplistic approach of LeprosyNet makes it the better option for solving specification problems with two categories.

$$\text{Precision} = \frac{\text{True positive}}{(\text{True positive} + \text{False positive})} \quad (1)$$

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$$\text{Recall} = \frac{\text{True positive}}{(\text{True positive}+\text{False negative})} \quad (2)$$

$$\text{F1} = \frac{2(\text{Precision}*\text{Recall})}{(\text{Precision}+\text{Recall})} \quad (3)$$

$$\text{Accuracy} = \frac{\text{True positive} + \text{True negative}}{(\text{True positive}+\text{True negative} + \text{False positive}+\text{False negative})} \quad (4)$$

Table 1: Model performance for each split

CNN Model	Precision	Recall	F1 Score	Accuracy
LeprosyNet	0.975	0.98	0.975	0.98
AlexNet	0.80	0.82	0.80	0.81
ResNet	0.80	0.80	0.80	0.80

	precision	recall	f1-score
Leprosy	0.99	0.97	0.98
Not Leprosy	0.96	0.99	0.97
accuracy			0.98
macro avg	0.97	0.98	0.98
weighted avg	0.98	0.98	0.98

FIGURE 8. Confusion matrix of LeprosyNet

	precision	recall	f1-score
0	0.84	0.81	0.82
1	0.76	0.79	0.77
accuracy			0.80
macro avg	0.80	0.80	0.80
weighted avg	0.80	0.80	0.80

FIGURE 9. Confusion matrix of ResNet

	precision	recall	f1-score
0	0.90	0.80	0.85
1	0.70	0.84	0.76
accuracy			0.81
macro avg	0.80	0.82	0.80
weighted avg	0.83	0.81	0.82

FIGURE 10. Confusion matrix of AlexNet

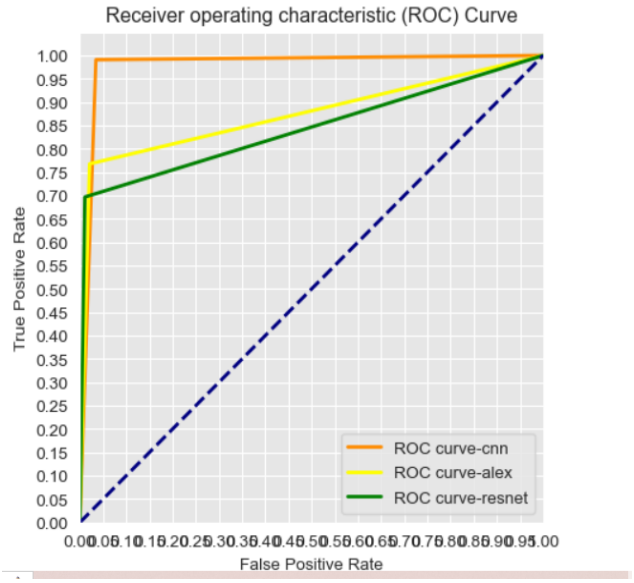


FIGURE 11. ROC curve of CNN Model

IX. COMPARISON WITH EXISTING RESEARCH

Table 2: Comparison of proposed system with other existing methods

References No	Classifiers used	Type of Dataset	Classification accuracy obtained (%)
8	Random Forest	EHR records (numerical data)	93.38%
23	CNN	Dermnetnz image-based dataset	91%
2	SVM	EHR records (numerical data)	98%
3	SVM	EHR records (numerical data)	98%
LeprosyNet	CNN	Dermnetnz image-based dataset	98%

Table 2 Compares the best proposed approach of this paper with existing research work. In [8] an application has been developed to combat leprosy using random forest technique with the accuracy of 93.38%, the data set used is based on parameters obtained with tedious contact-based techniques which is an obvious bottleneck.

In [22] the authors have proposed a CNN architecture for classification using images from Dermnetnz dataset. They have achieved average accuracy of 91% which has a scope of improvement. CNN is black box technique, and the features on which predictions are being made are not clearly determined.

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In [2] and [3] Authors have used machine learning techniques like logistic regression and SVM to classify leprosy into different categories. In both the papers the best accuracy achieved is 98% using SVM whereas logistic regression has shown 95% accuracy. This classification has been undertaken with the help of Electronic Health Records of ailing patients. These EHR are based on complex contact-based techniques for detection which even though accurate, seldom detect cases at a later stage. In the future scope of [3] it is recommended to study leprosy based on an image-based dataset using CNN.

Thus, the obvious advantages of LeprosyNet over existing architectures are that it provides a contact-less mode of detection which has successfully passed rigorous XAI based reliability tests.

X. CONCLUSION AND FUTURE WORKS

The best proposed LeprosyNet architecture showed an average accuracy of 98% on the test set. Its performance is better than state-of-art architectures like AlexNet and RestNet, It has also shown better results than existing image based leprosy detectors. Its performance is comparable to EHR based detectors with the added advantage of noninvasive, contact-less prediction. By the results obtained from XAI techniques, it is evident that the model can segregate the lesion affected area from the redundant parts of the image. This is an added proof of the credibility of predictions made by the model. In future this model can be integrated into an app or website and can be used as a preliminary Leprosy detection test.

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